

Software Design Report
CSC 221: Programming 2: Spring 2024
Dr. Bareiss
Fourth Lab
Tiago Silva

Problem Summary

Create a GUI environment for the Tic Tac Toe program

Implementation Requirement

- Must include a complete UML diagram that accurately represents the software design, incorporating major functions such as game components and their interactions.
- Must provide a detailed interface drawing that identifies various GUI components and their respective locations within the software.
- Must emphasize testing on the GUI and integration aspects of the software, acknowledging that the game components are already functional.
- Must have Default size (3x3) with player names input.
- Must Read game setup information from a file.
- Must have only one class is permitted to perform direct I/O.

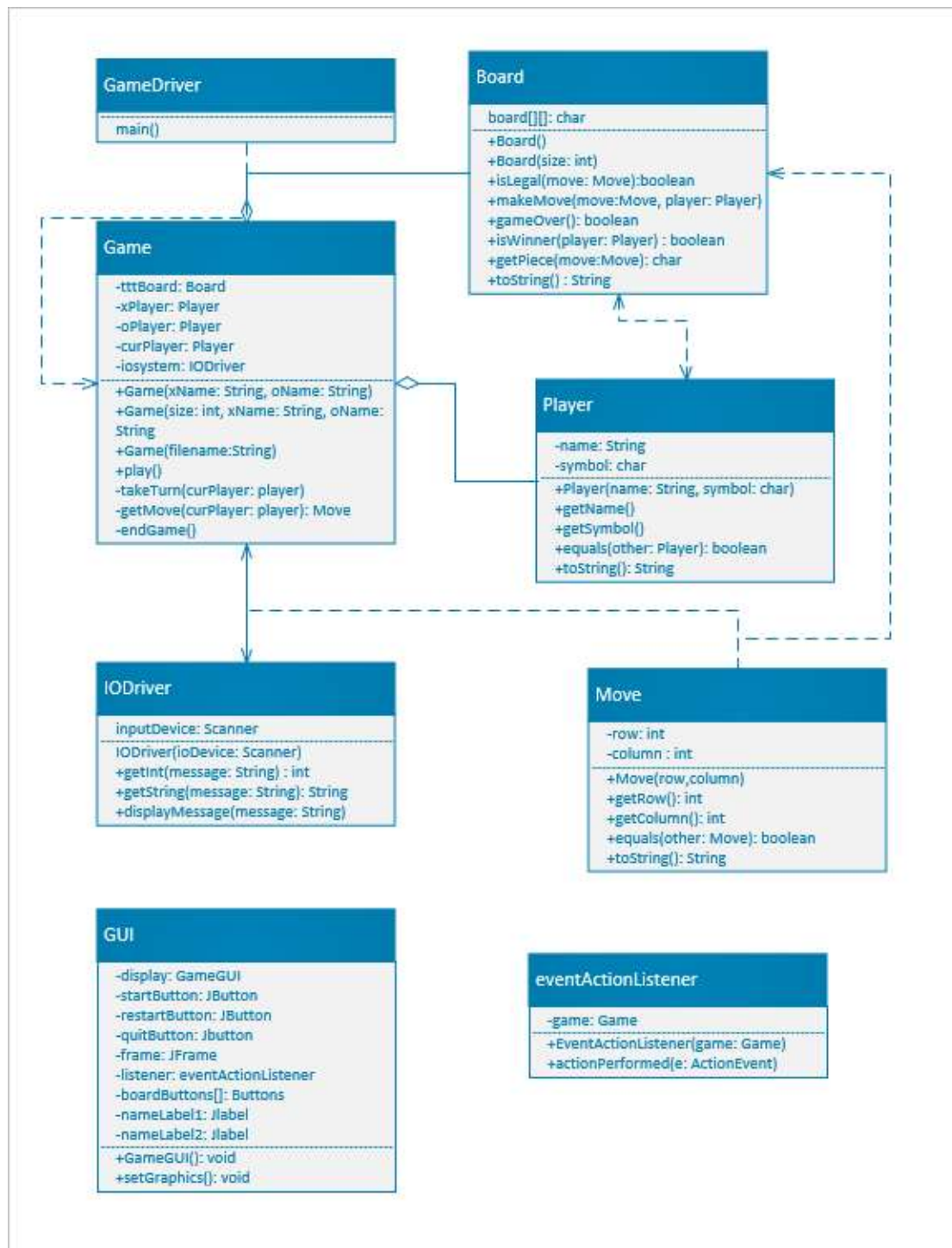
Testing

Class	Name	Description of test	Code	Expected Output	Actual Output
			Click menu bar button		
GUI	Start Game	Test starting the game	Click the start game option	Starts the game with the names and the board size chosen	
			Click menu bar button		
	End Game	Test ending the game	Click the end game option	The game ends and the program closes	

	File Input	Test using a file to load a game	Give the name of the file and run the game	Loads the game from the file	
	Names	Test giving the names and the size	Run program	(size chosen) Board with the names displayed in the bottom corners	
			Input the names and board size.	The user that is playing will show up on top of the game board	
	Size smaller than 3	Test using a size smaller than 3	Run Program	Game default size = 3x3	
			Test numbers lower than 3		
			Run program		
	Use Button Multiple times	Test using a button more than once	Click a button, after use disable it		
			Click the same button again	Button disabled	
			Run Program		
	Size of board string	Use text instead of na integer	Input the names		
			Board size: "seven"	Game default size = 3x3	
	Winning Cases	Test if the winning cases are working	X X O X O O X O X	X wins (Vertical)	
			X O O		

			O O X		
			X O X	O wins (Vertical)	
			X O X		
			O X X		
			O O X	X wins (Vertical)	
			O O O		
			O X X		
			X O X	O Wins (Horizontal)	
			X O O		
			X X X		
			X O O	X wins (Horizontal)	
			X O O		
			O X O		
			X X X	X wins (Horizontal)	
			O O X		
			O X X		
			X O X	X wins (Diagonal)	
			O O X		
			X O X		
			X O O	O wins (Diagonal)	

System design



Time Spent Estimative

4h total for code, testing, and SDR.

Outside Resources used

Professor's Code

Potential security risks

The program collects and uses player names, which could raise privacy concerns if the data is not handled securely.

Potential ethical implications

The game should be accessible to all users, including those with disabilities. Considerations should be made for providing alternative input methods or accommodation to ensure everyone can enjoy the game.

Lesson Learned

Learned how to test properly and how to test more efficiently.

Not everything in programming is coding; there's many other roles and testing is one of them. I don't love it, but it's an important part of programming.

Improvements of Work

I fixed my UML diagram and some small details on the testing plan